

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

NET207-R

Network cost optimization on AWS

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AWS

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(he/him)

Principal Solutions Architect

AWS



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What you will learn in this session

- Fundamentals of network cost
- Recent pricing changes
- Q&A

Fundamentals



Networking pricing components

Fixed component (\$/hour)

AWS Transit Gateway, NAT gateway, Amazon Route 53, AWS PrivateLink endpoints, Elastic Load Balancing (ELB), AWS Direct Connect, AWS VPN, AWS Cloud WAN, AWS Global Accelerator, Amazon VPC Lattice, Amazon VPC IP Address Manager (IPAM), and so on

Examples

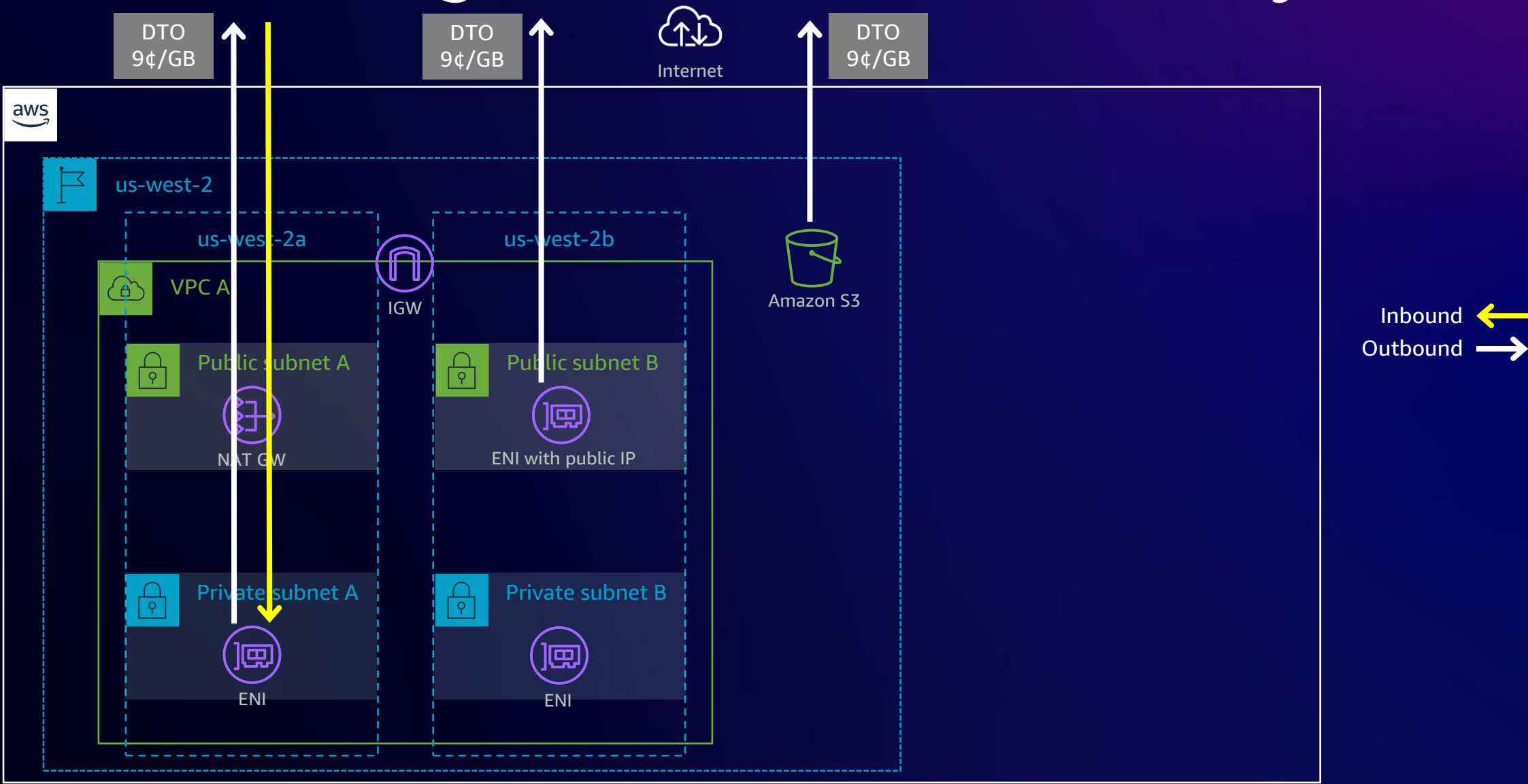
- 10 Gbps Direct Connect in US = $\$2.25/\text{hour} \times 730 \text{ hours/month} = \$1,642.5 \text{ per month}$
- Multi-AZ NAT gateway = $\$0.045/\text{hour} \times 730 \text{ hours/month} \times 2 = \65.7 per month
- 10 Transit Gateway VPC attachments = $\$0.05/\text{hour} \times 730 \text{ hours/month} \times 10 = \365 per month

Variable component (\$/GB or \$/requests)

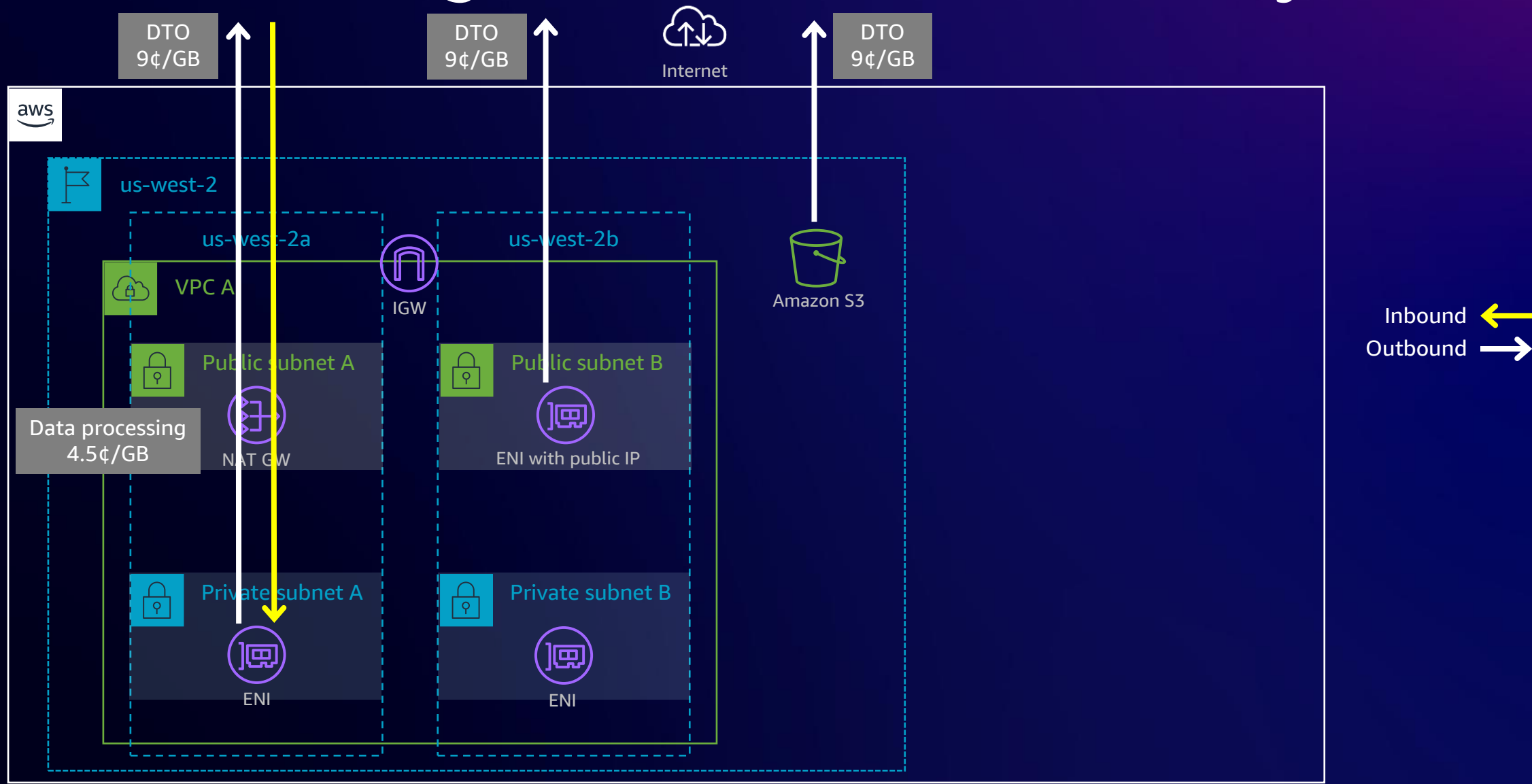
- Data transfer – DTO, DTAZ, Inter-Region
- Data processing – Transit Gateway, NAT gateway, PrivateLink, AWS Cloud WAN, VPC Lattice
- DNS queries or CloudFront HTTP/HTTPS requests



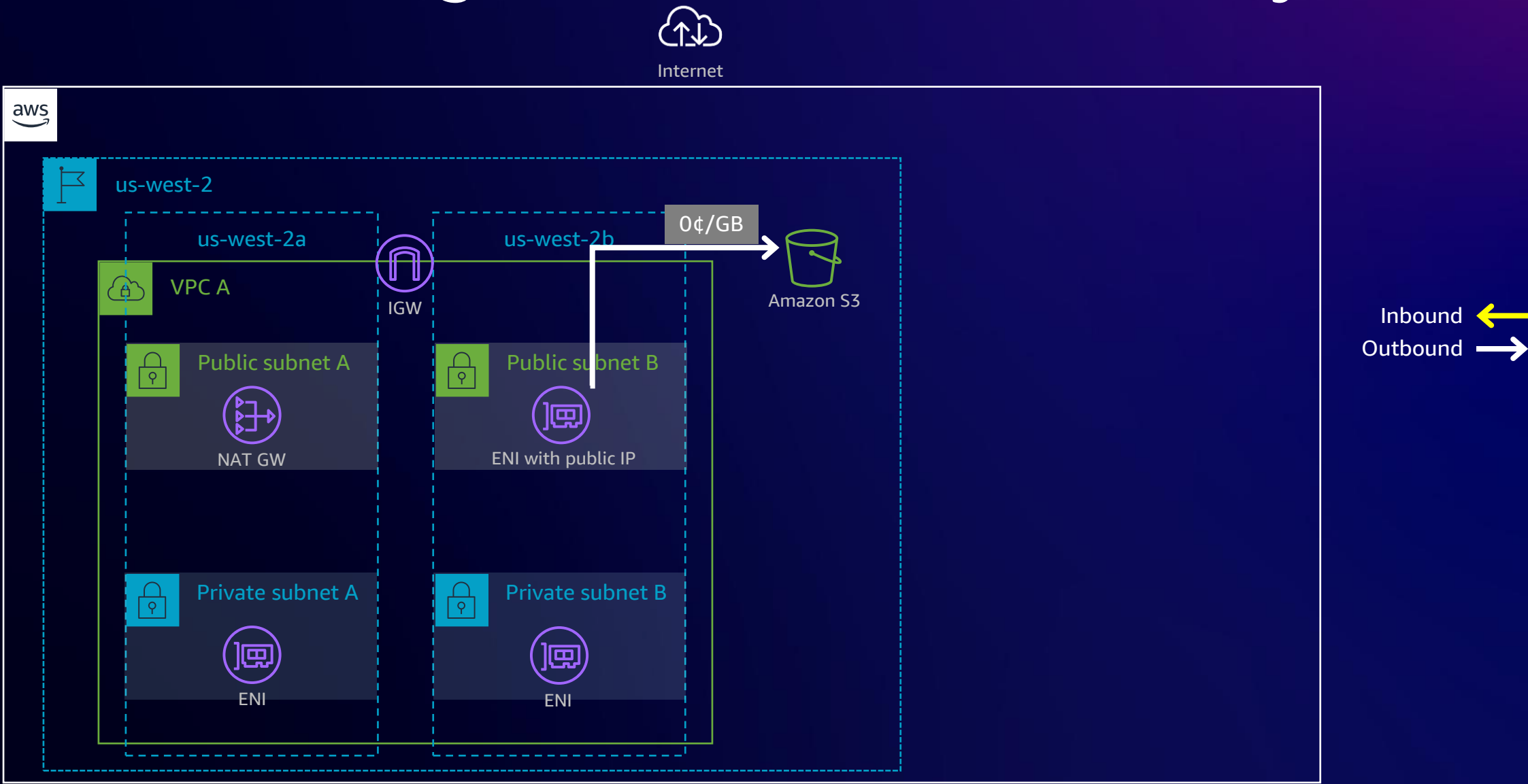
Internet, inter-Region, and inter-Availability Zones



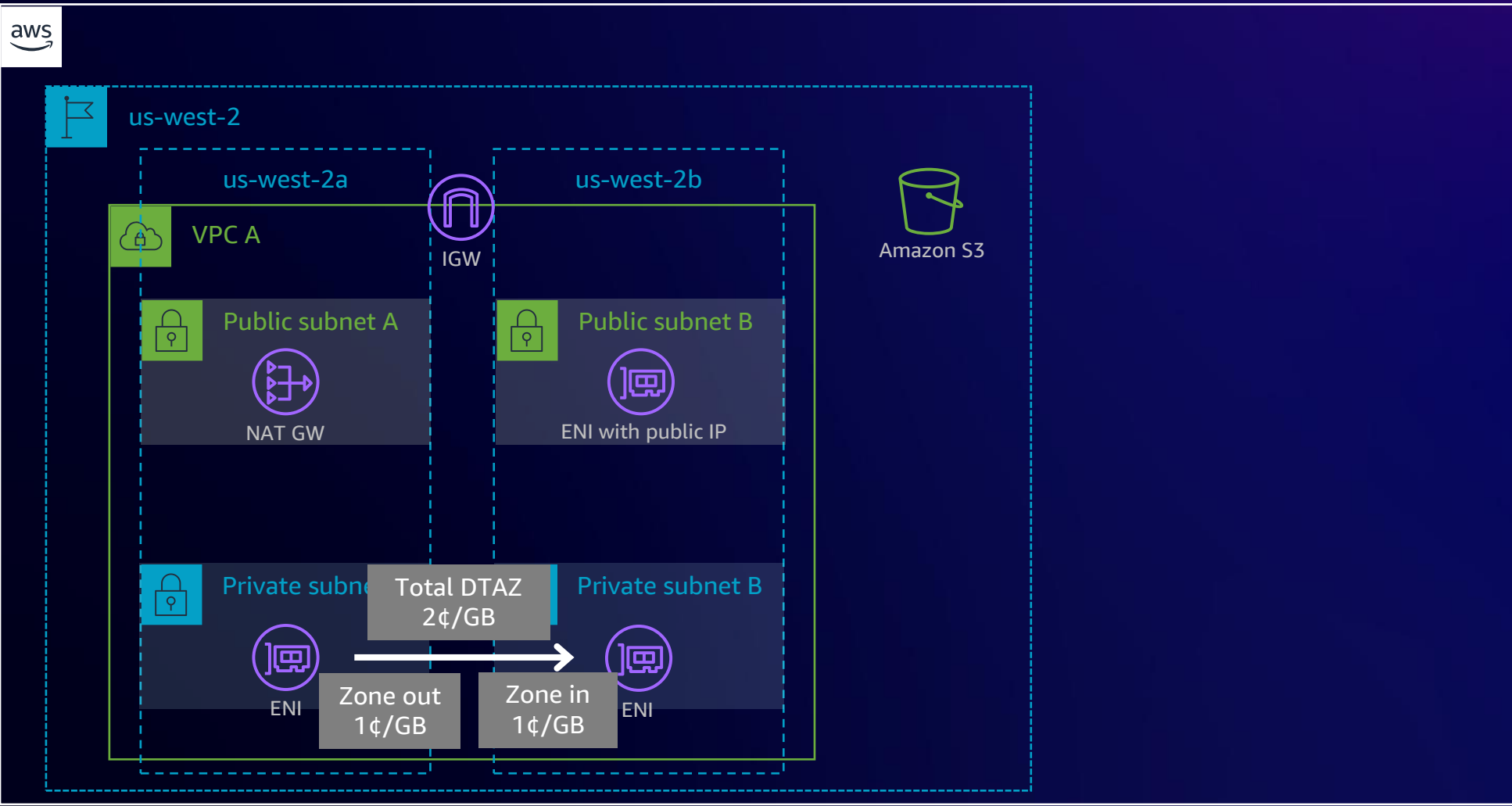
Internet, inter-Region, and inter-Availability Zones



Internet, inter-Region, and inter-Availability Zones



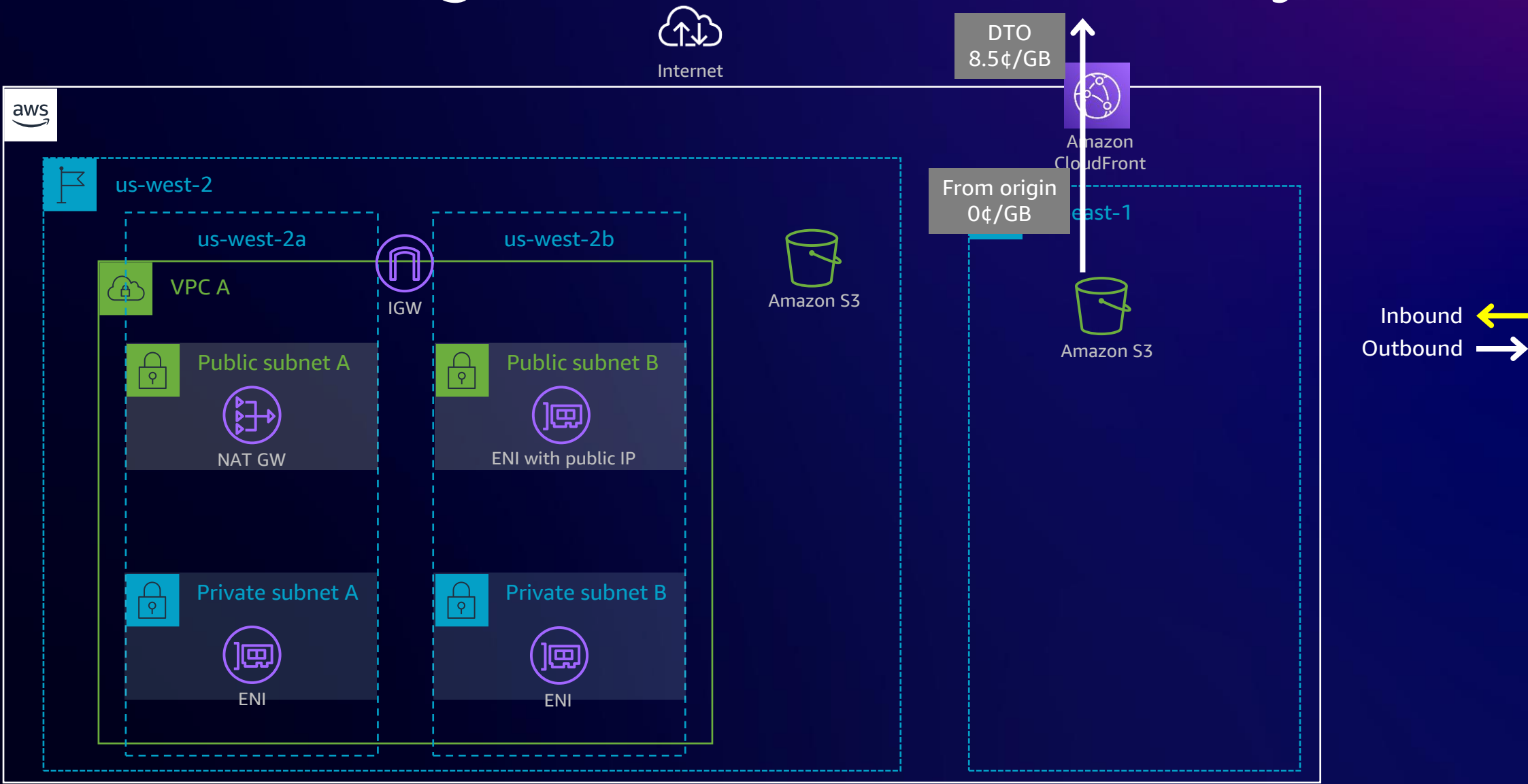
Internet, inter-Region, and inter-Availability Zones



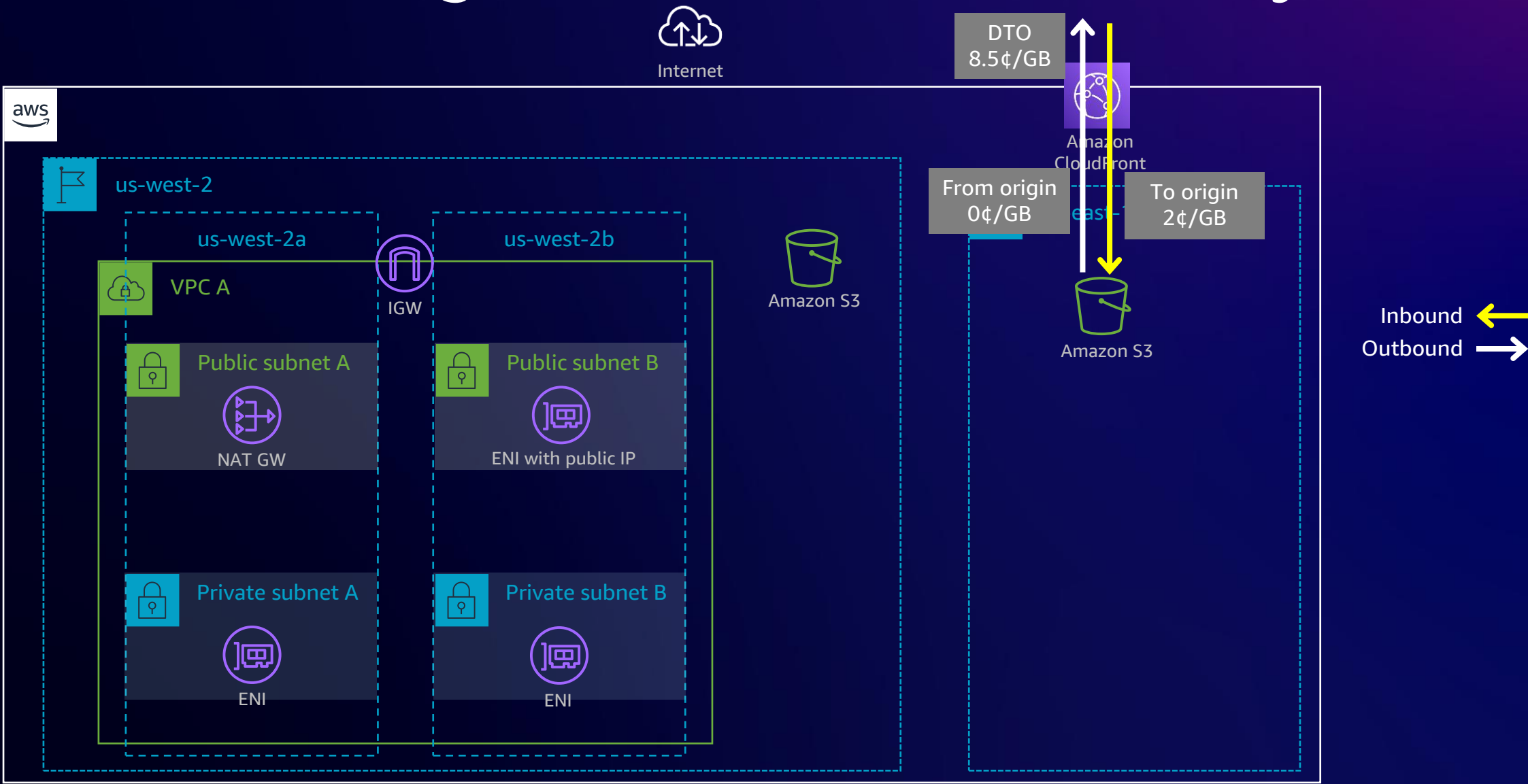
Inbound ←
Outbound →



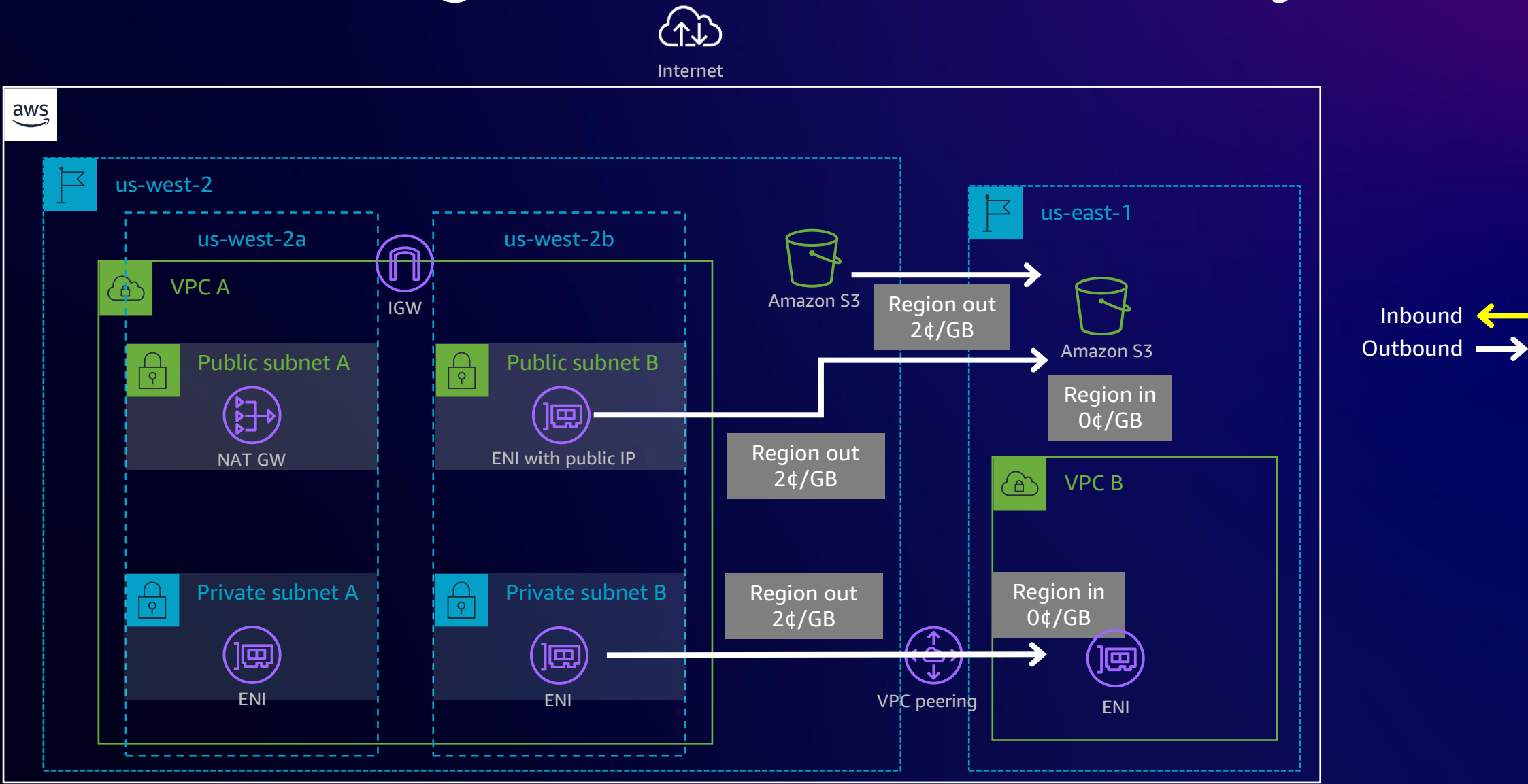
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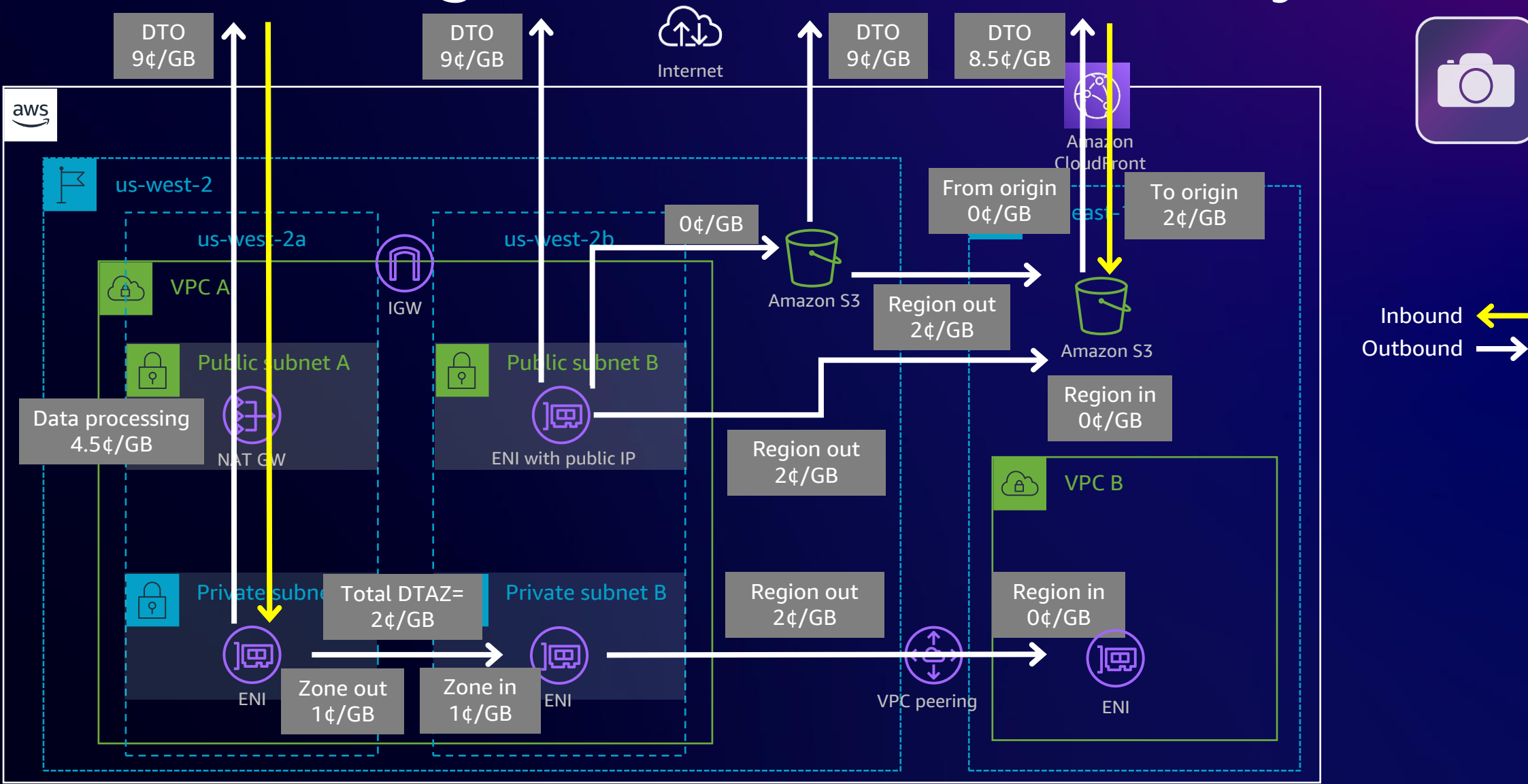
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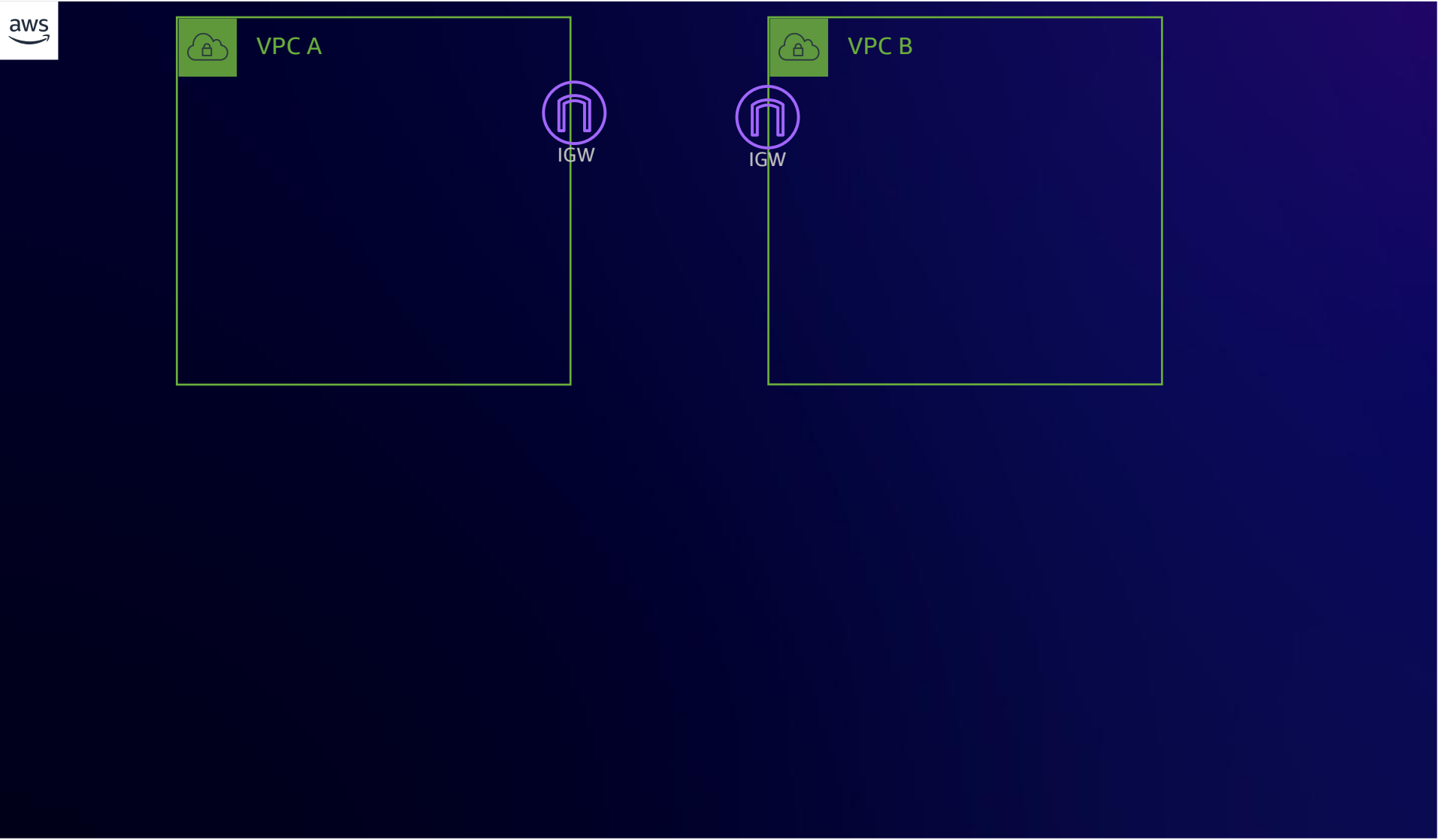
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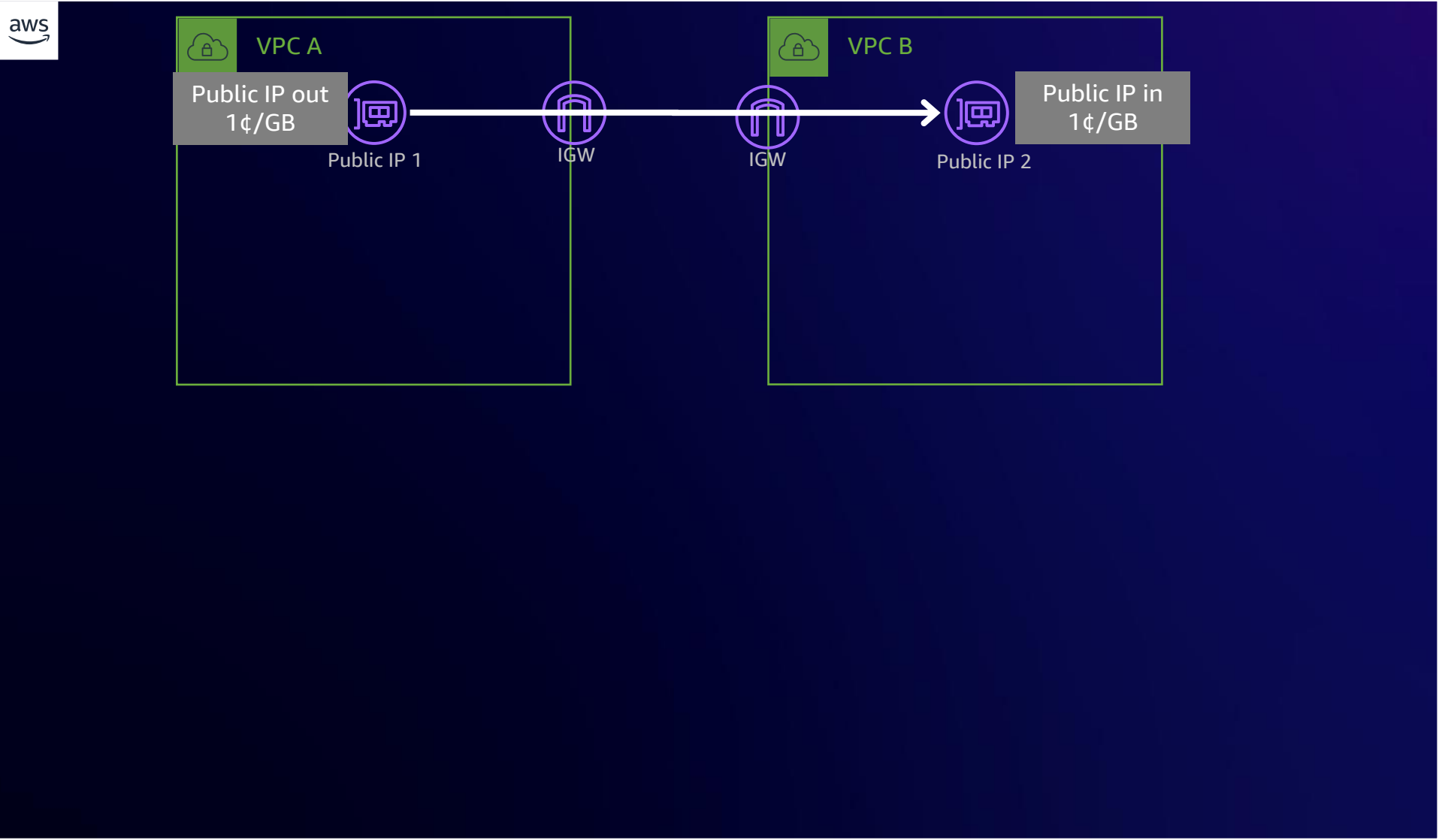


Inter-VPC, endpoints, and hybrid



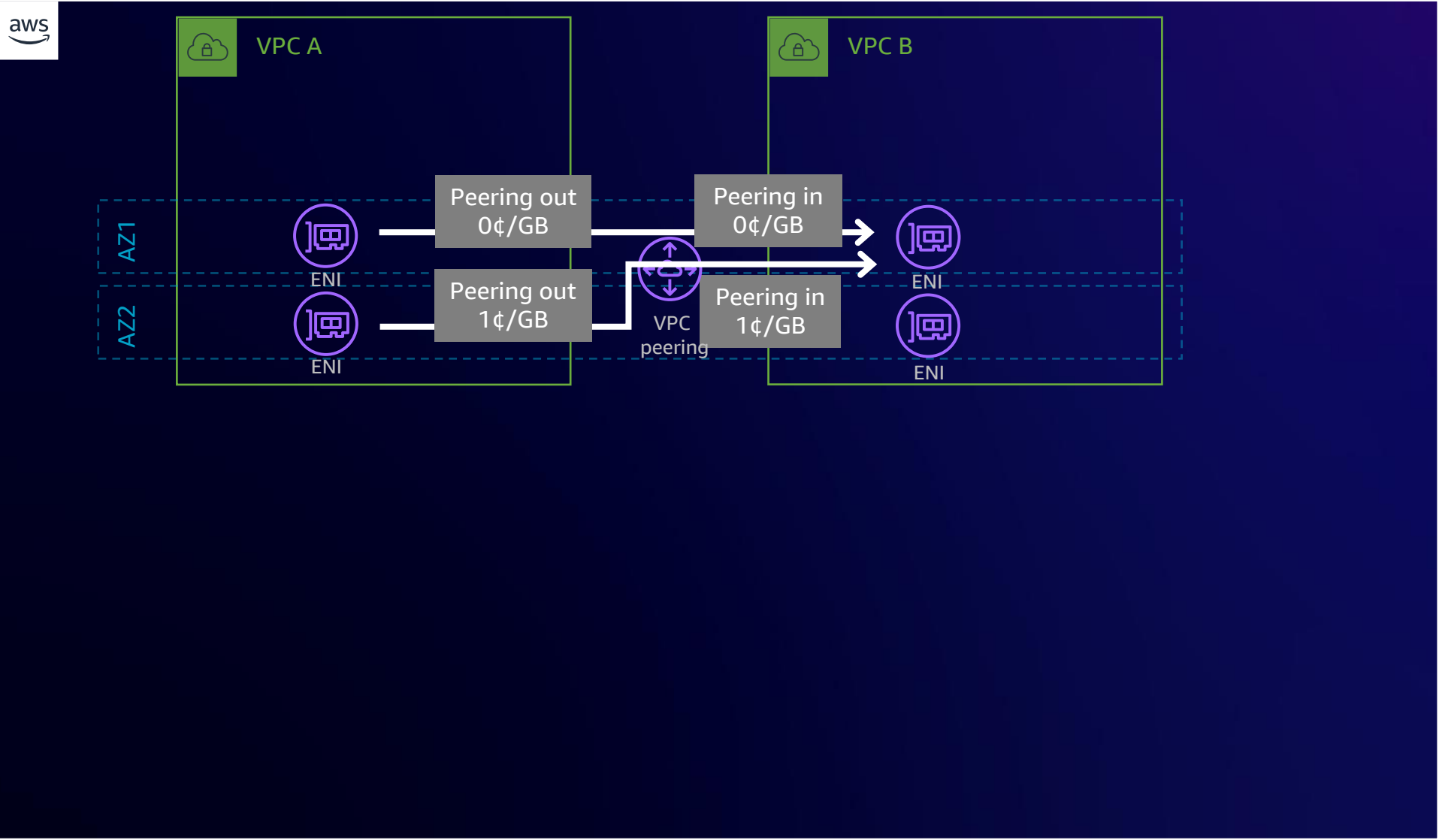
Inbound ←
Outbound →

Inter-VPC, endpoints, and hybrid

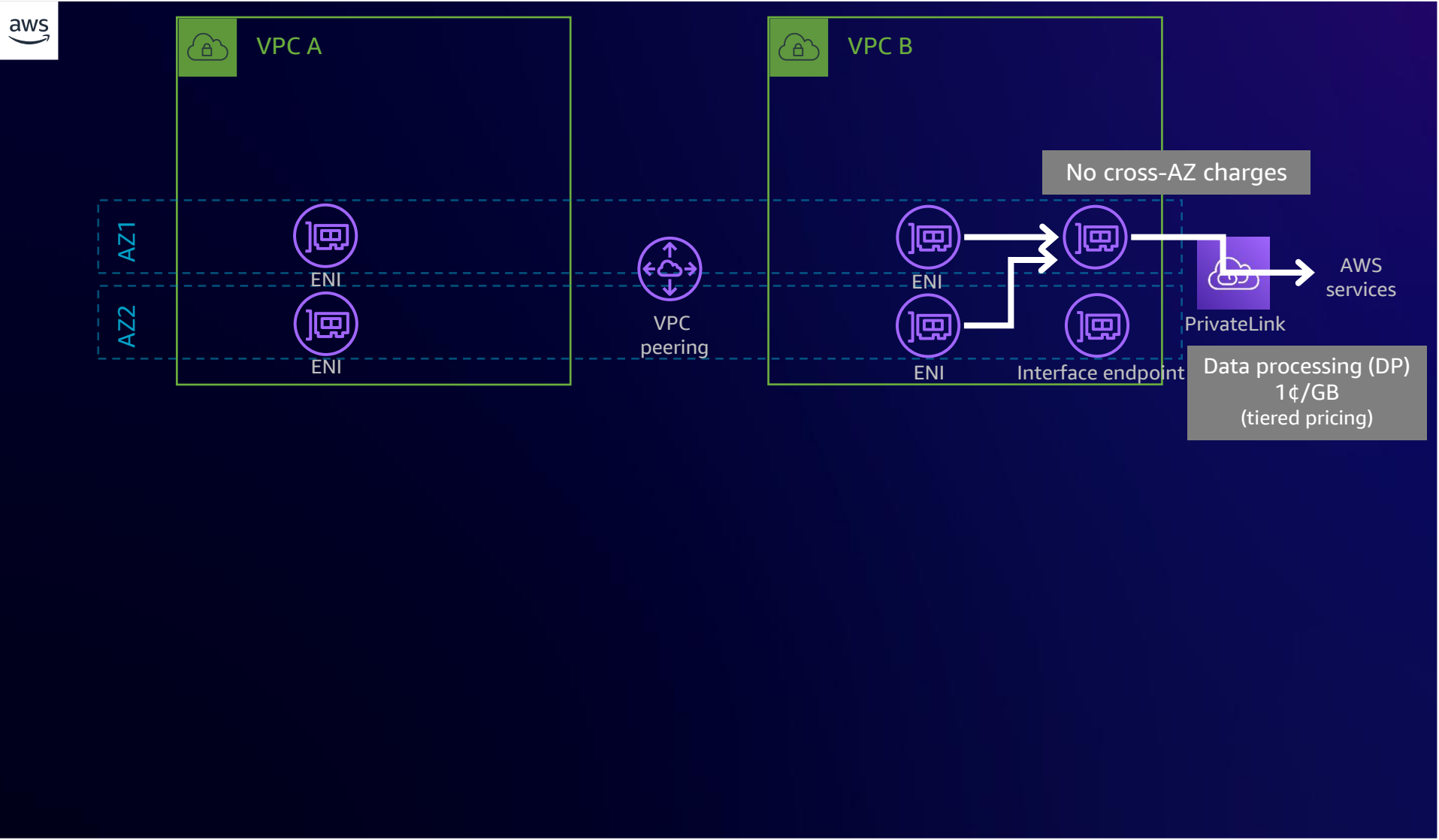


Inbound ←
Outbound →

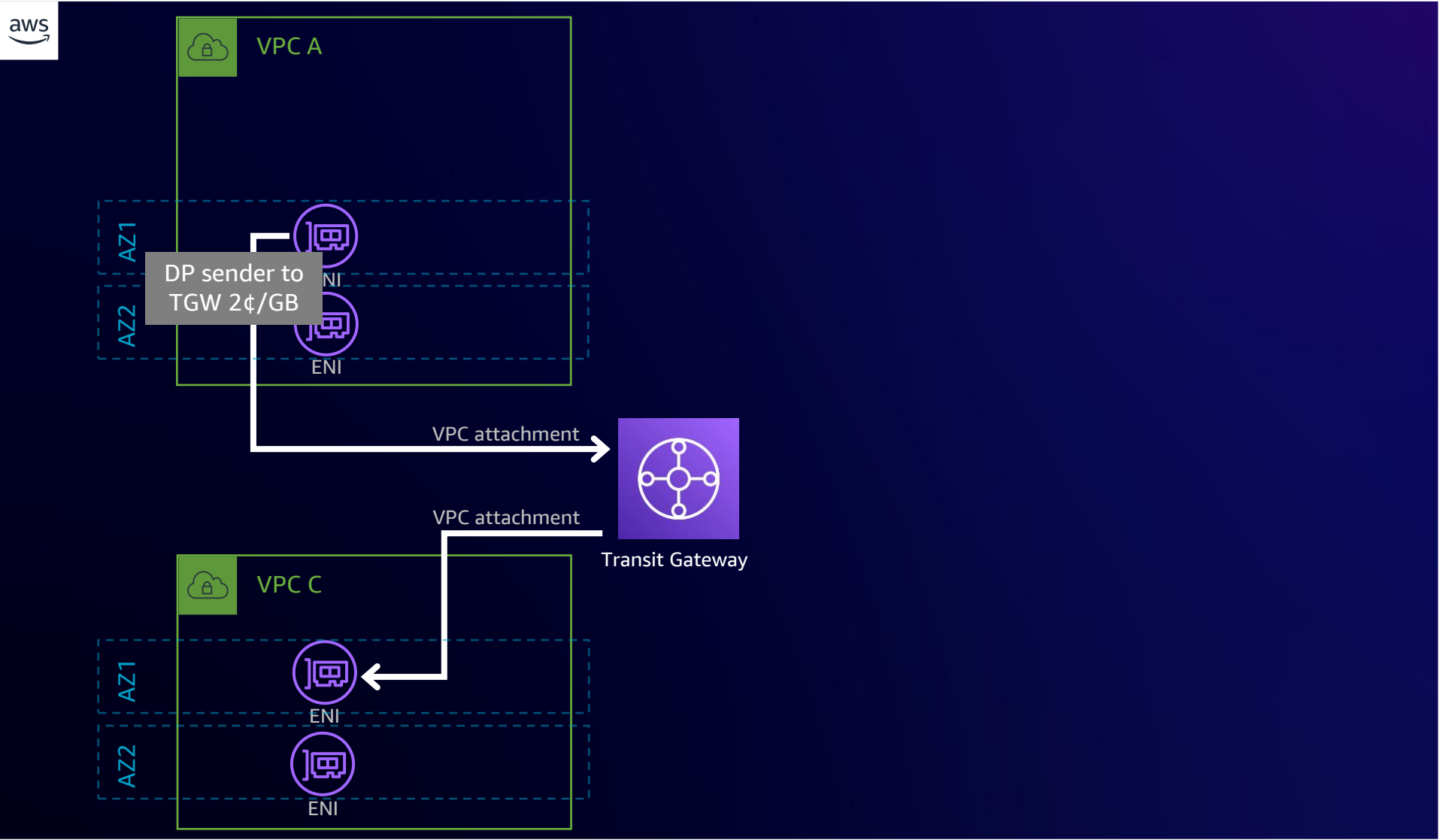
Inter-VPC, endpoints, and hybrid



Inter-VPC, endpoints, and hybrid

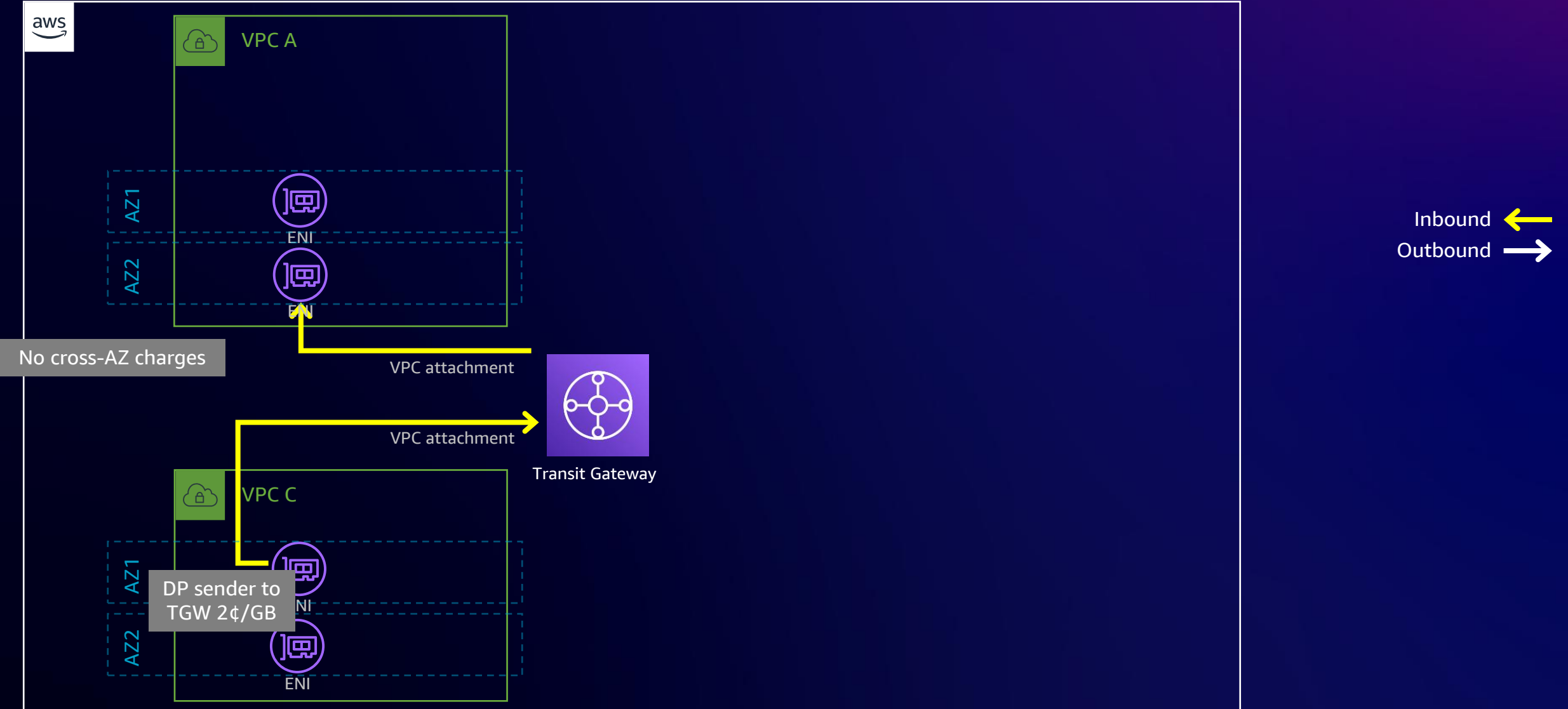


Inter-VPC, endpoints, and hybrid

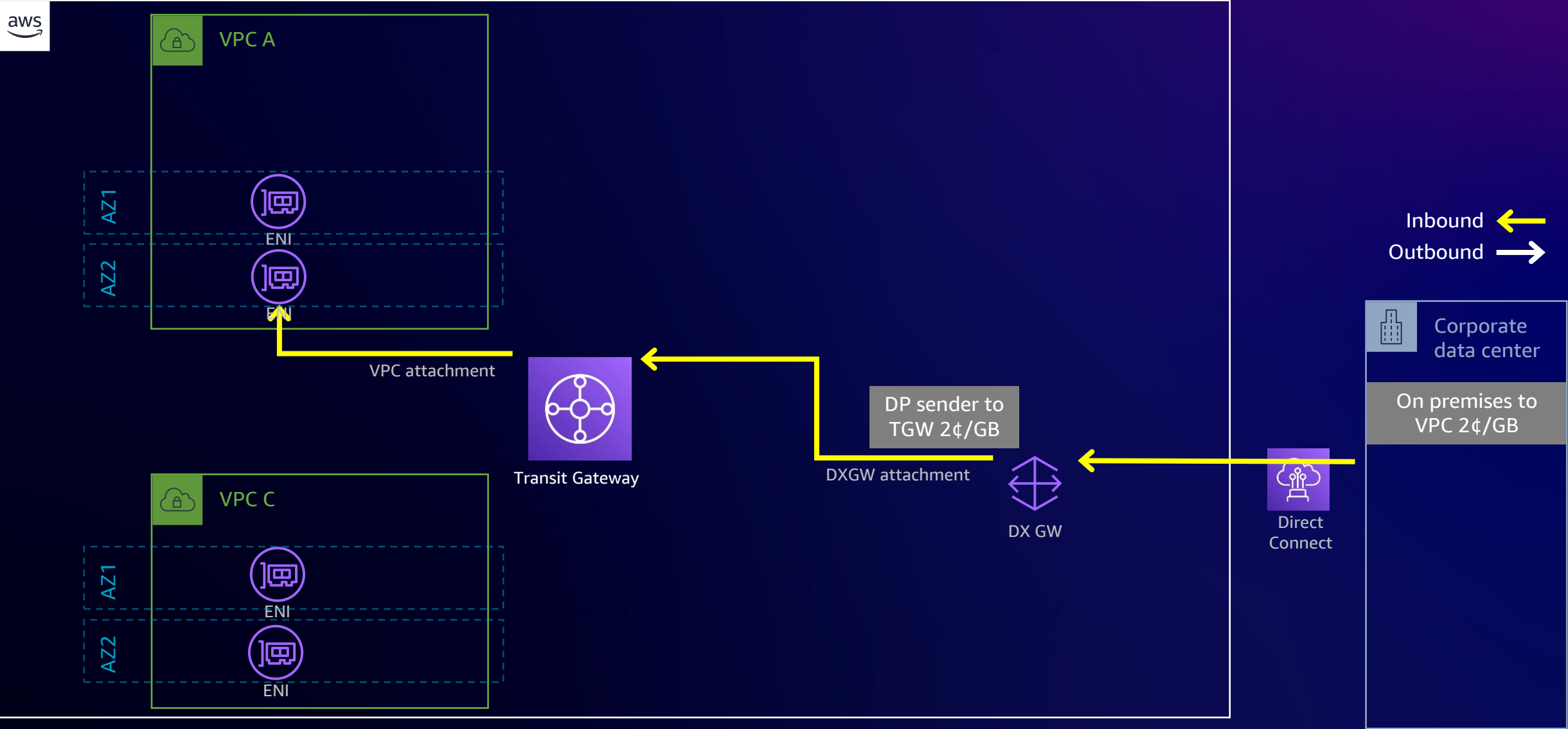


Inbound ←
Outbound →

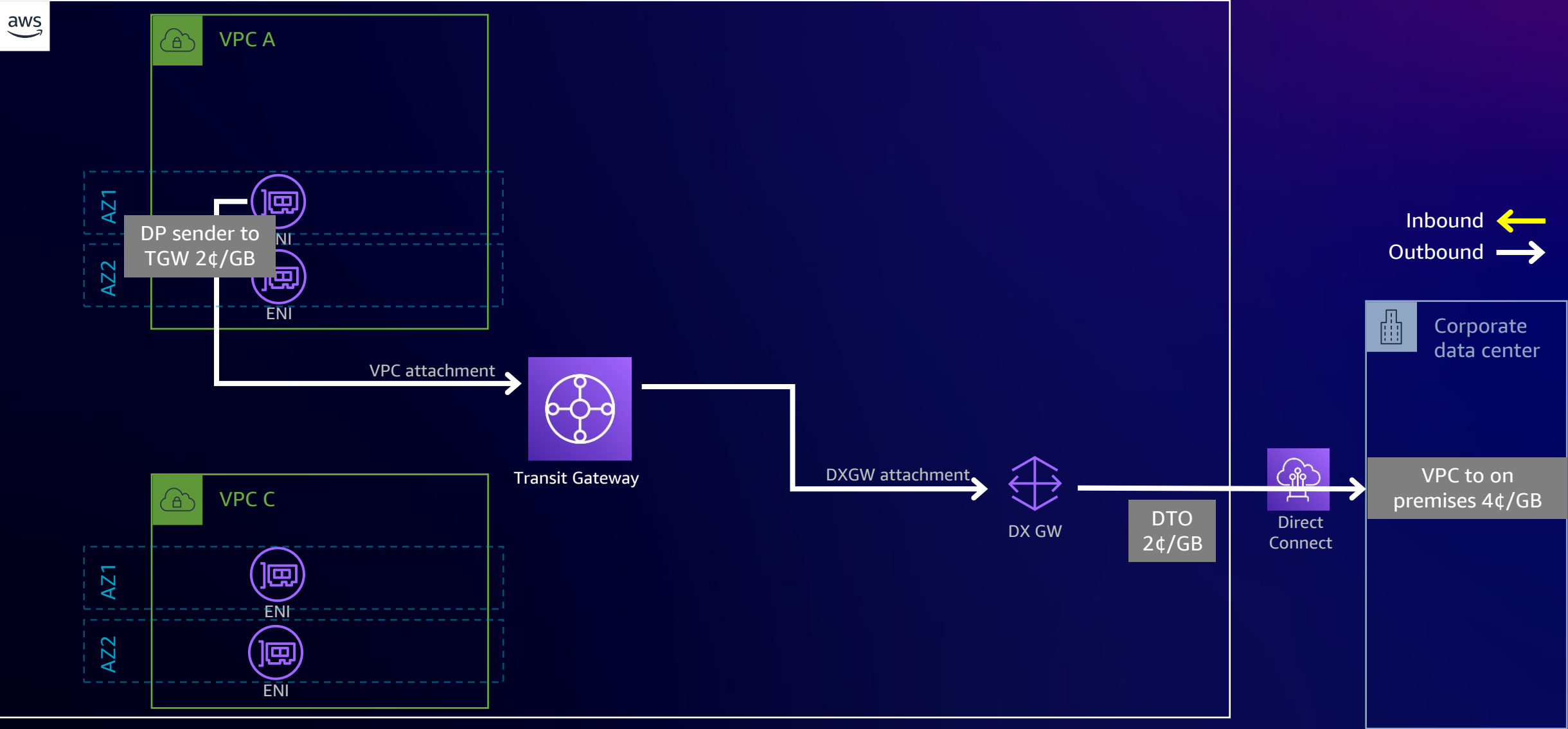
Inter-VPC, endpoints, and hybrid



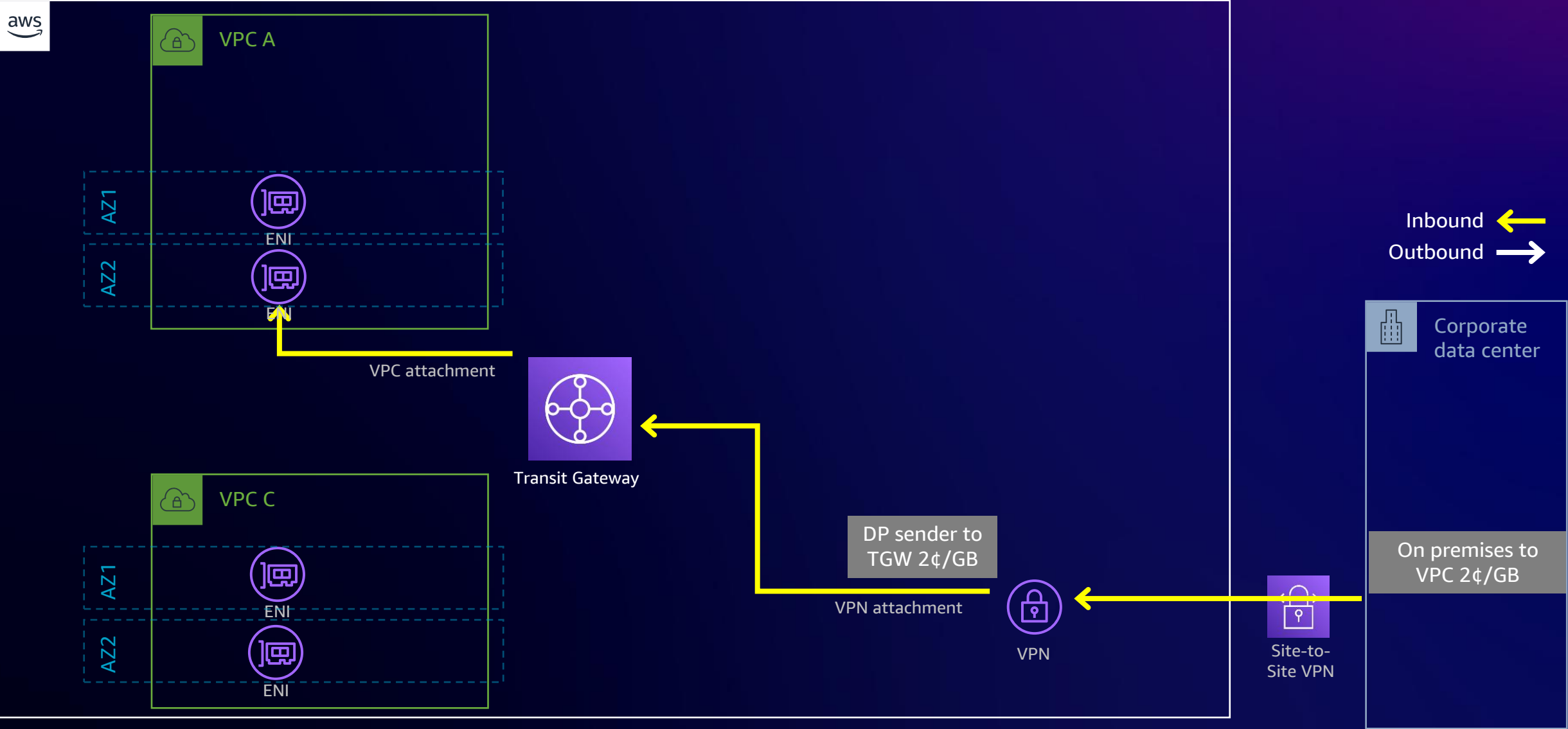
Inter-VPC, endpoints, and hybrid



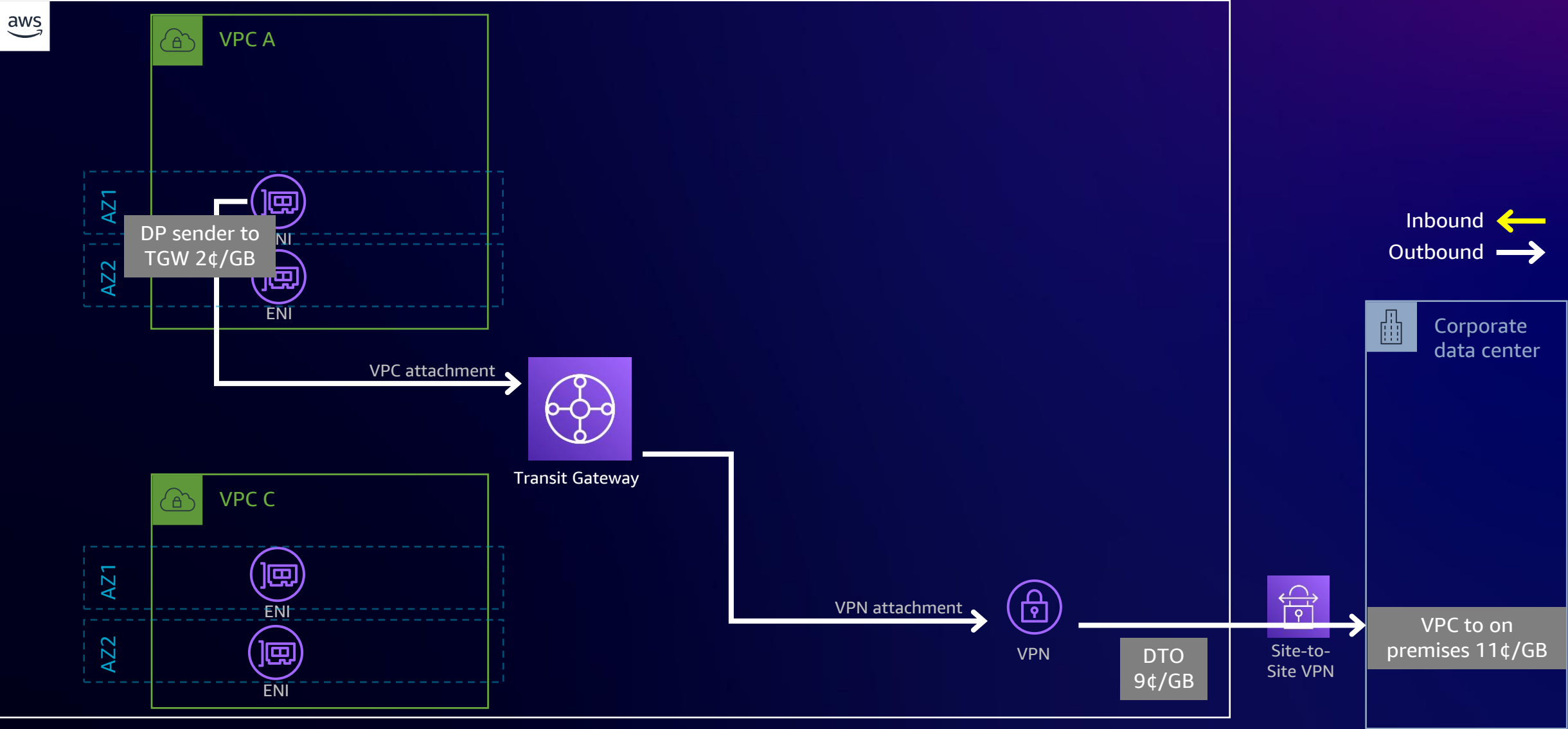
Inter-VPC, endpoints, and hybrid



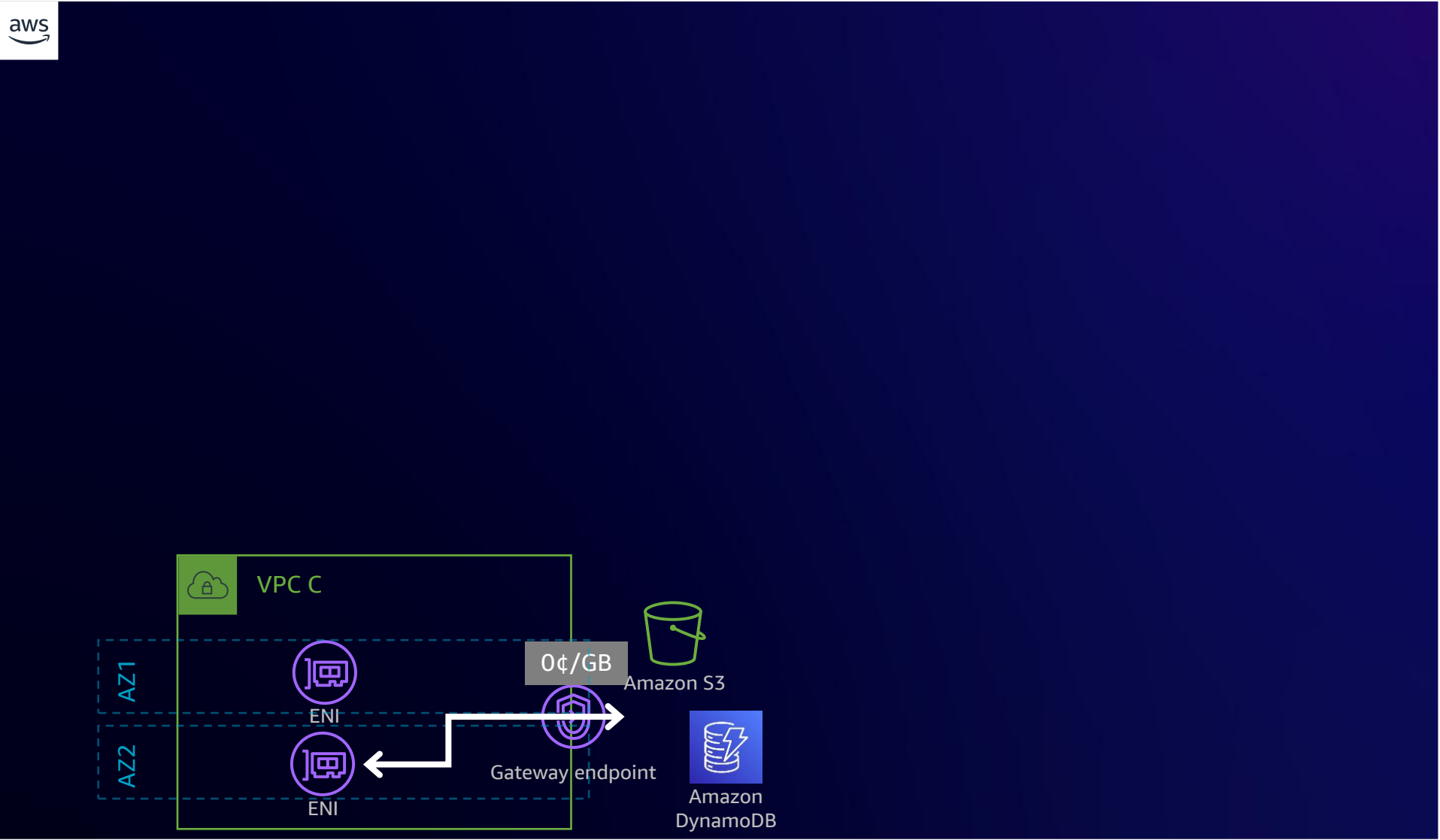
Inter-VPC, endpoints, and hybrid



Inter-VPC, endpoints, and hybrid

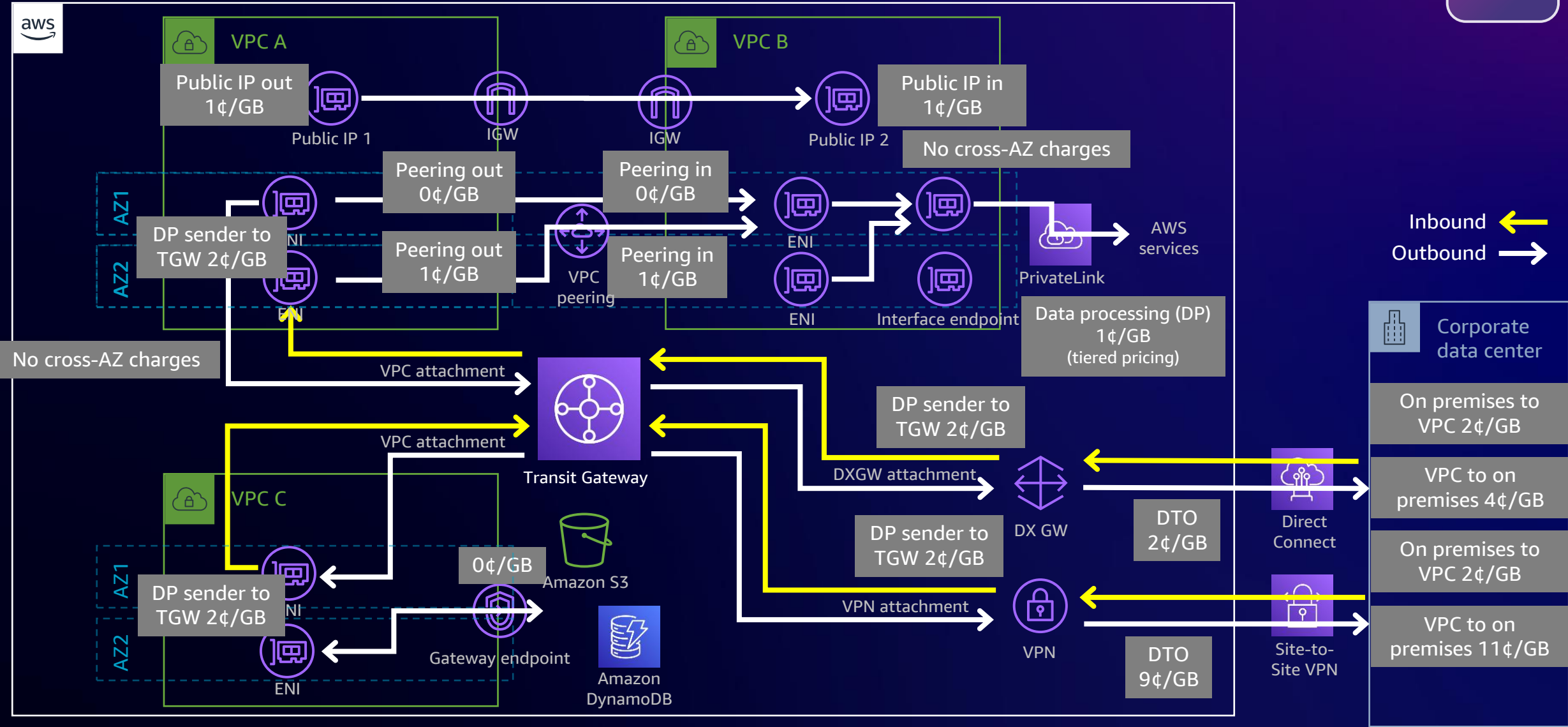


Inter-VPC, endpoints, and hybrid



Inbound ←
Outbound →

Inter-VPC, endpoints, and hybrid



Recent pricing changes



Few changes since last year



- VPC peering has no data transfer charges (except inter-AZ)
- No inter-AZ charges for TGW, VPN, PrivateLink scenarios
- Public IP pricing effective 02/01/2024



<https://aws.amazon.com/blogs/networking-and-content-delivery/identify-and-optimize-public-ipv4-address-usage-on-aws/>

Key takeaways

- Networking cost dimensions – fixed vs. variable
- How data transfer and data processing charges differ
- How network cost works across common connectivity options
- Estimate data transfer costs by baseline traffic traversing your AWS network

Q&A



Thank you!



Please complete the session
survey in the mobile app

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Sohaib Tahir

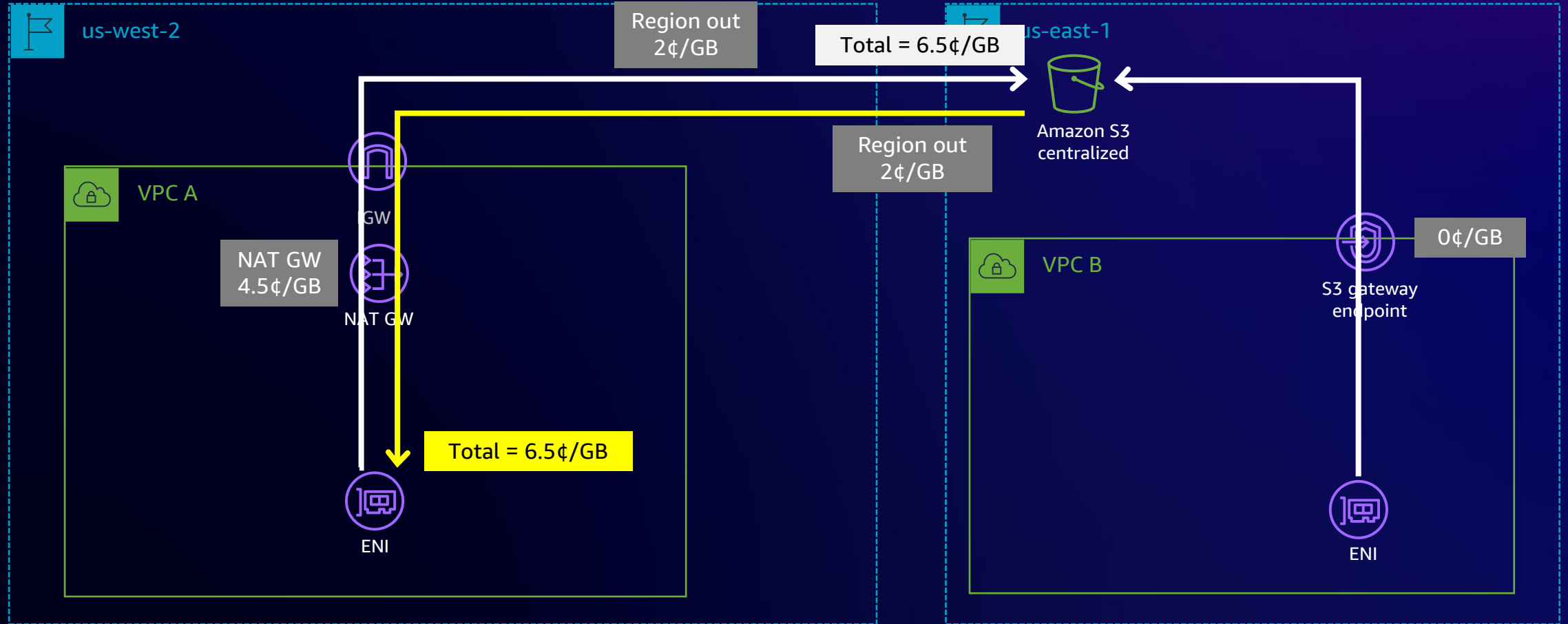
[linkedin.com/in/syed-sohaib-tahir-046b0915](https://www.linkedin.com/in/syed-sohaib-tahir-046b0915)



Appendix

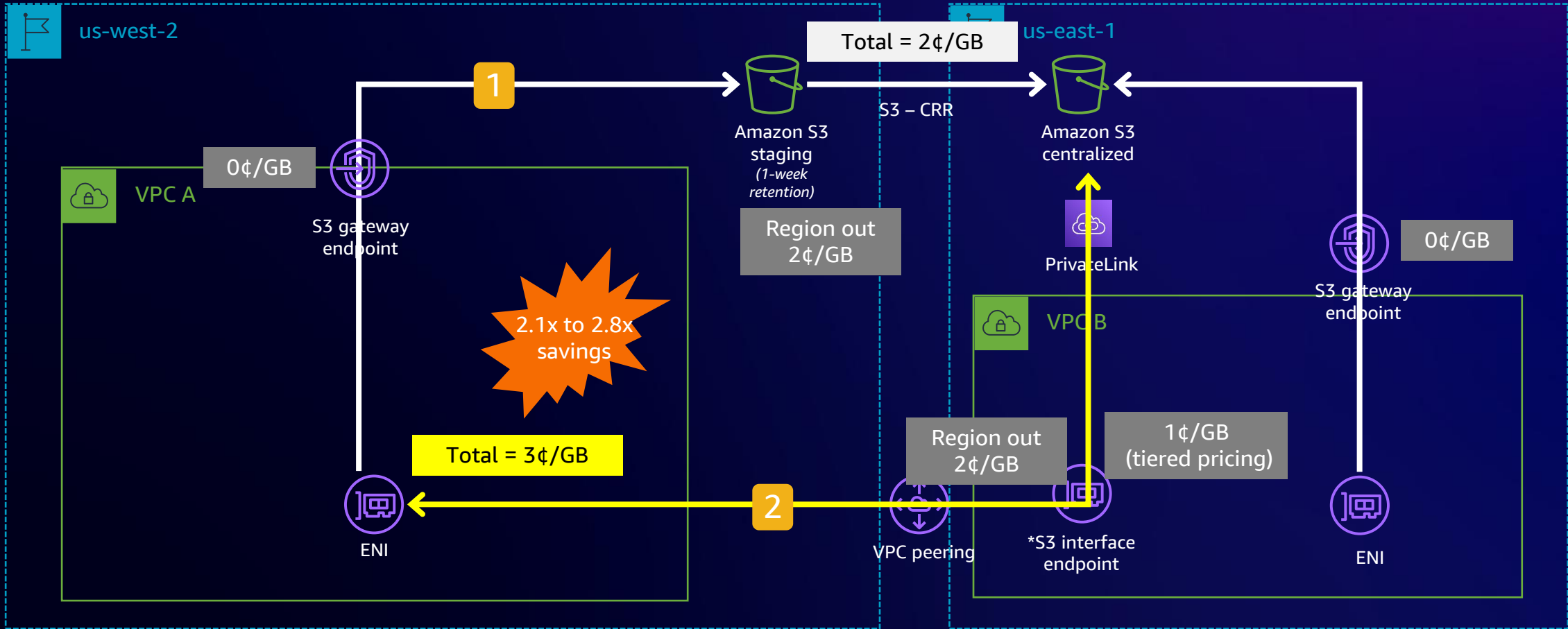
Scenario #1 – Centralized S3 bucket (non-optimized)

PROBLEM: NAT GW PROCESSING CHARGES ARE 80% OF DATA TRANSFER



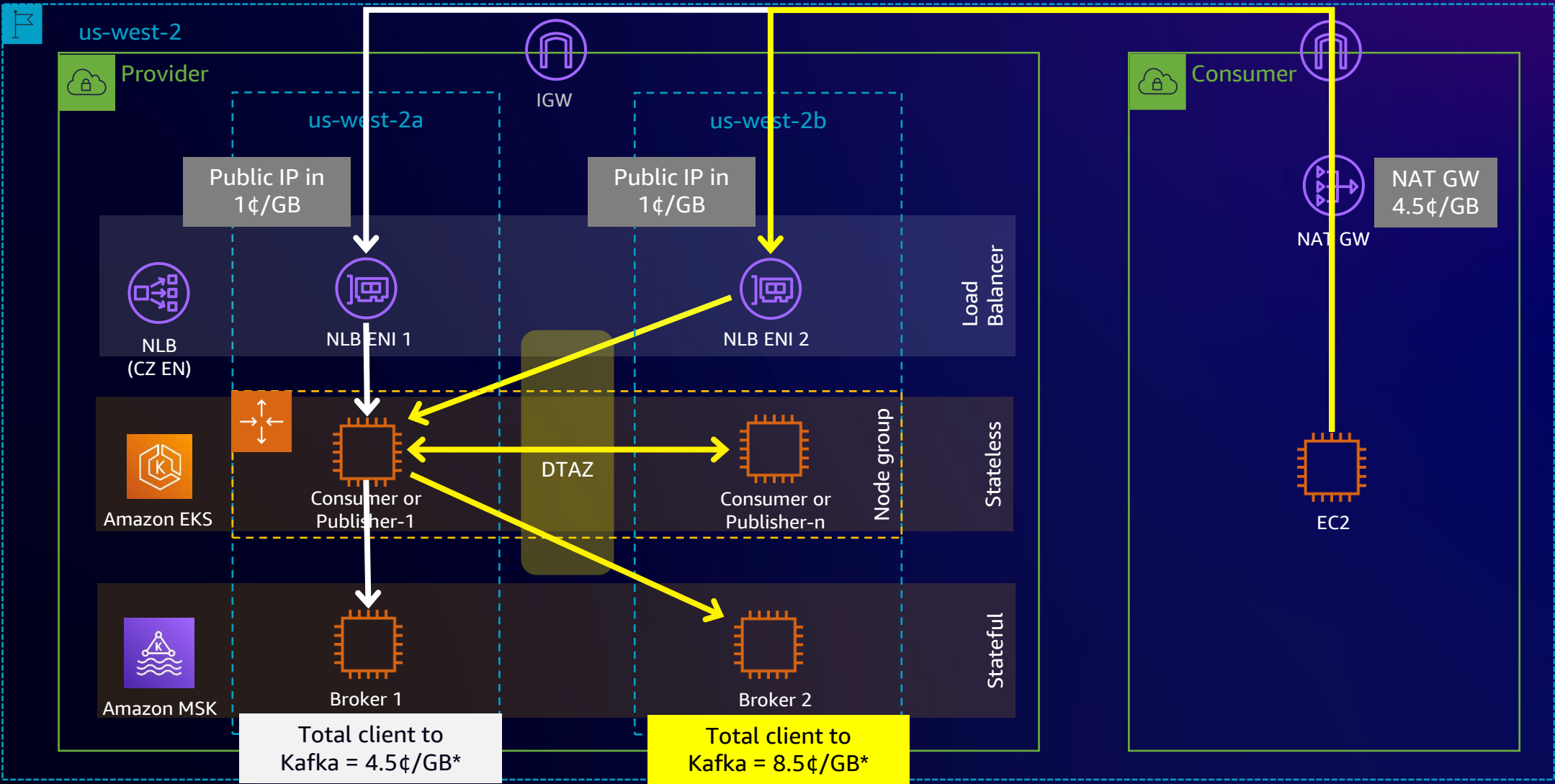
Scenario #1 – Centralized S3 bucket (optimized)

SOLUTION: REDUCE NAT GW COST BY DIVERTING REGION-OUT TRAFFIC THROUGH ENDPOINTS



Scenario #2 – Amazon EKS and Amazon MSK (non-optimized)

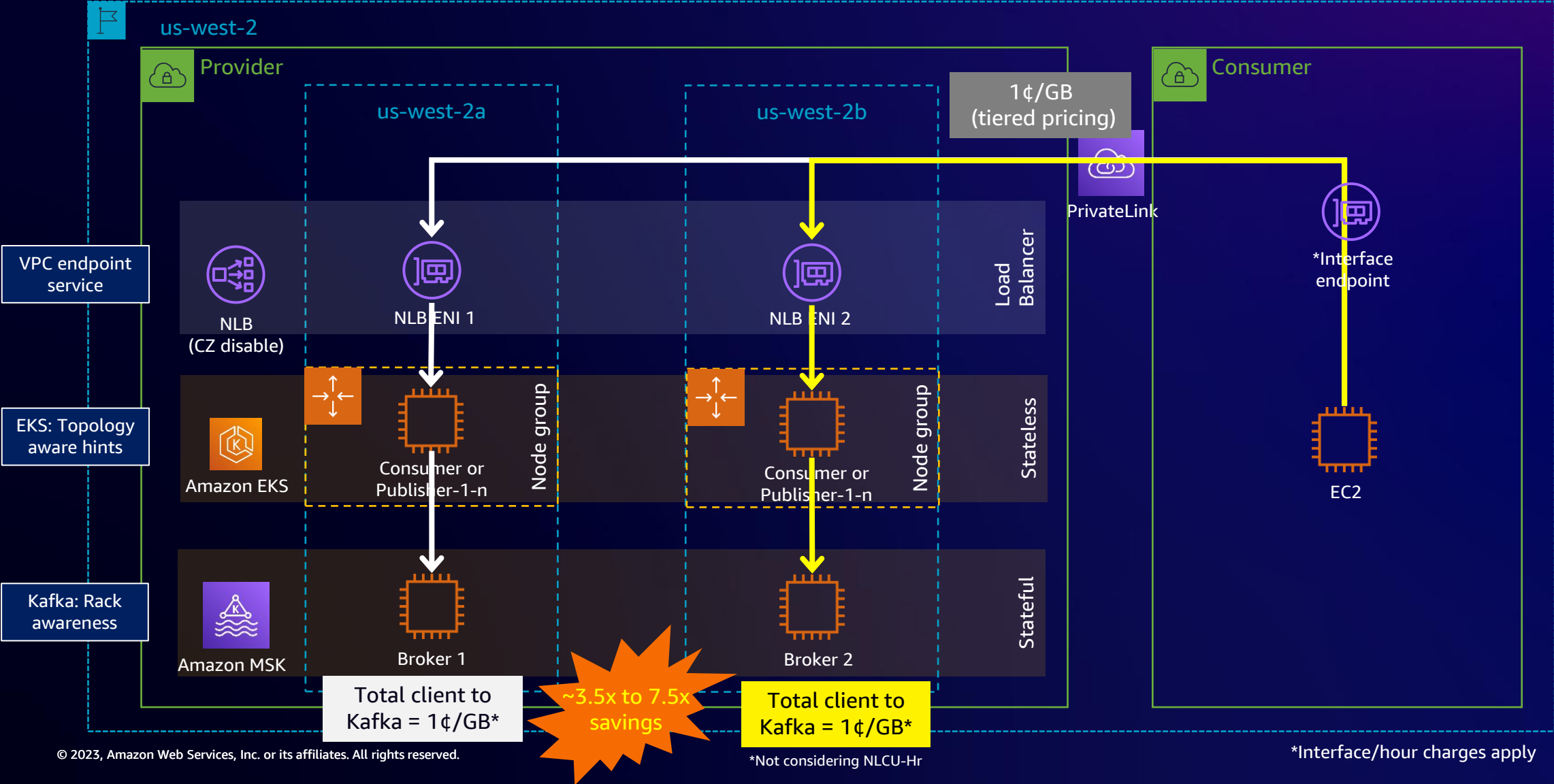
PROBLEM: 70% OF DATA TRANSFER CHARGES ARE DUE TO INTER-AZ (REGIONAL DATA TRANSFER)



*Not considering inter-AZ charges incurred by Spark or Aerospike nodes and NLCU-Hr

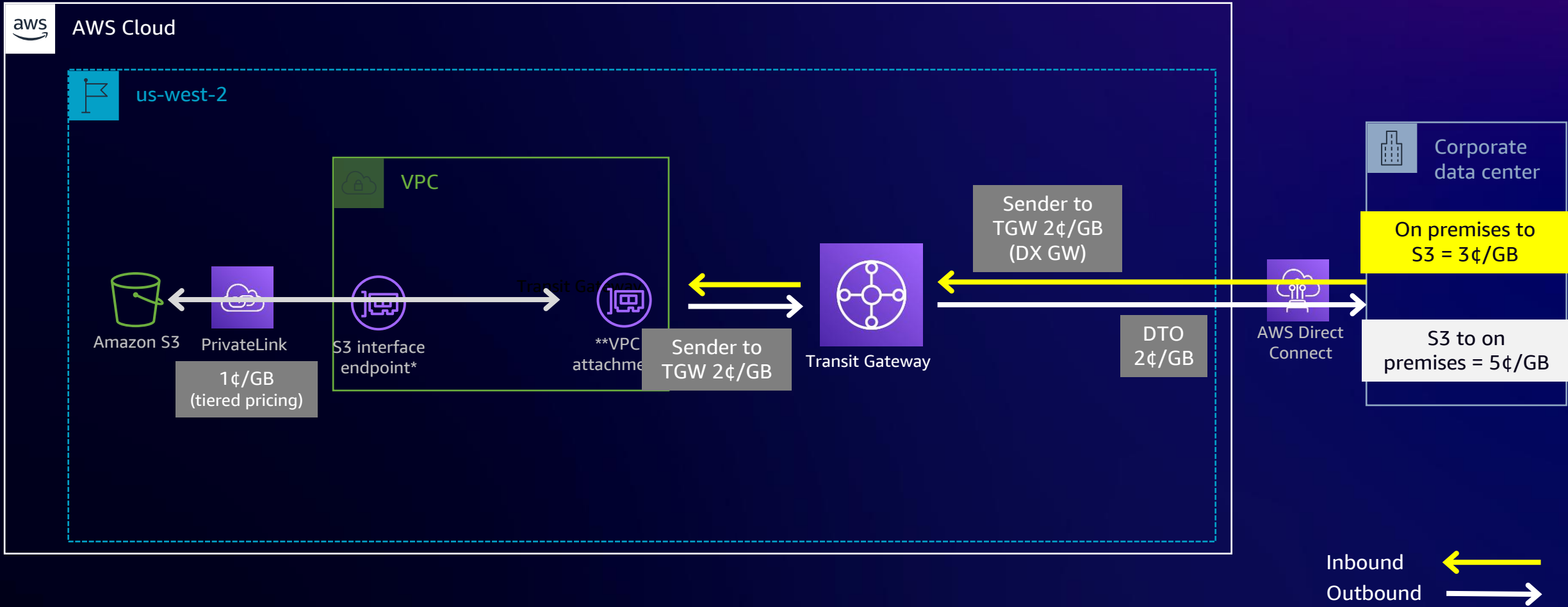
Scenario #2 – Amazon EKS and Amazon MSK (optimized)

SOLUTION: ENABLE ZONAL AWARENESS TO REDUCE INTER-AZ CHARGES



Scenario #3 – Amazon S3 private connectivity (non-optimized)

PROBLEM: HIGH TGW PROCESSING CHARGES



Scenario #3 – Amazon S3 private connectivity (optimized)

SOLUTION: USE VGW FOR S3 INTERFACE ENDPOINTS TO REDUCE DATA PROCESSING CHARGES

